

PROJECT PART-B

AIM -CONSTRUCT AND SIMULATE A NOR GATE USING MOS TRANSISTORS

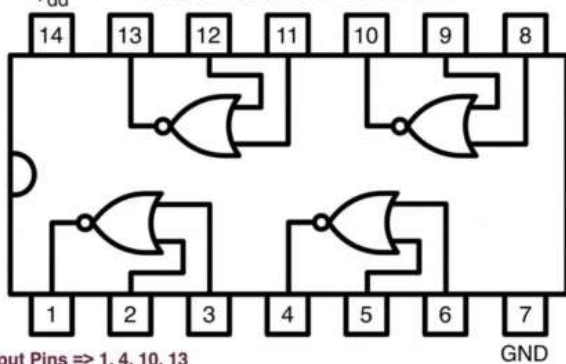
NOR Gate



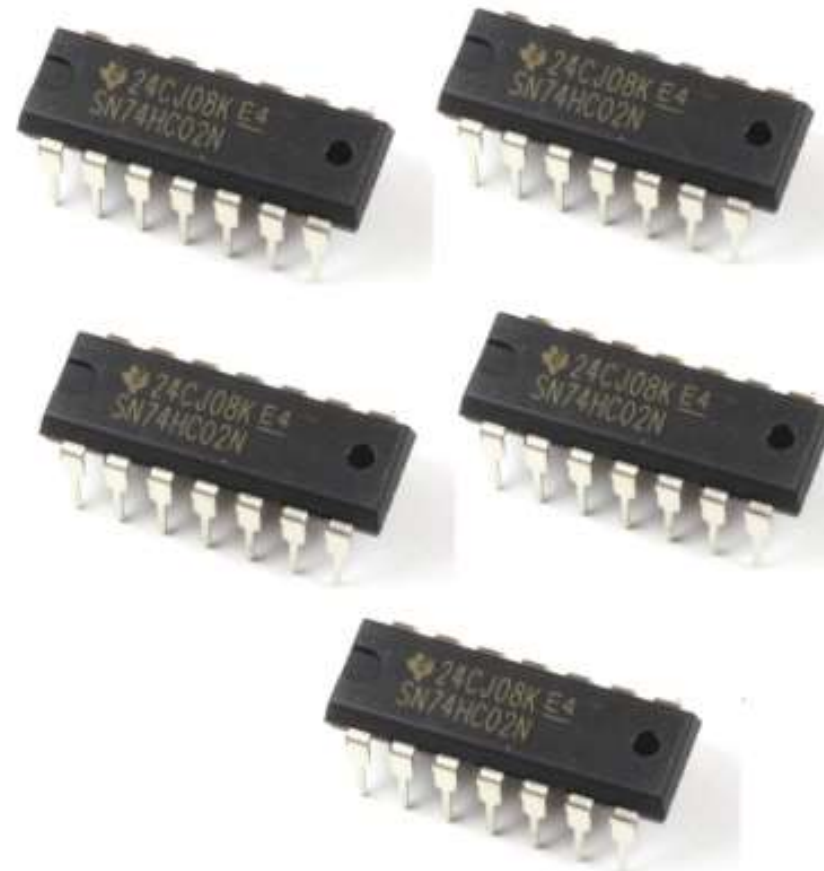
Truth Table

A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

NOR Gate IC 7402



- Output Pins => 1, 4, 10, 13
- Input Pins => (2, 3), (5, 6), (8, 9), (11, 12)

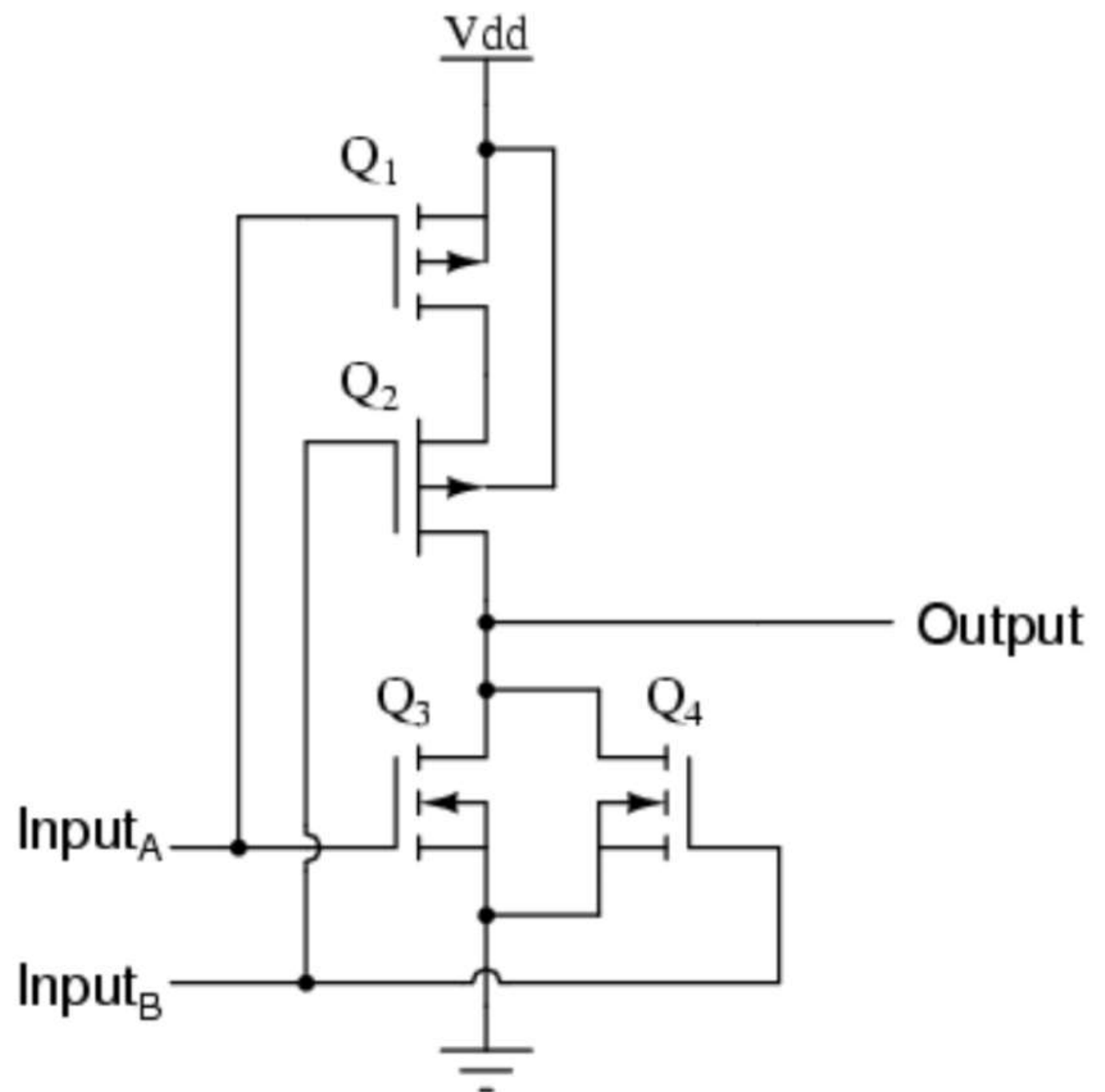


COMPONENTS USE TO MAKE NOR GATE

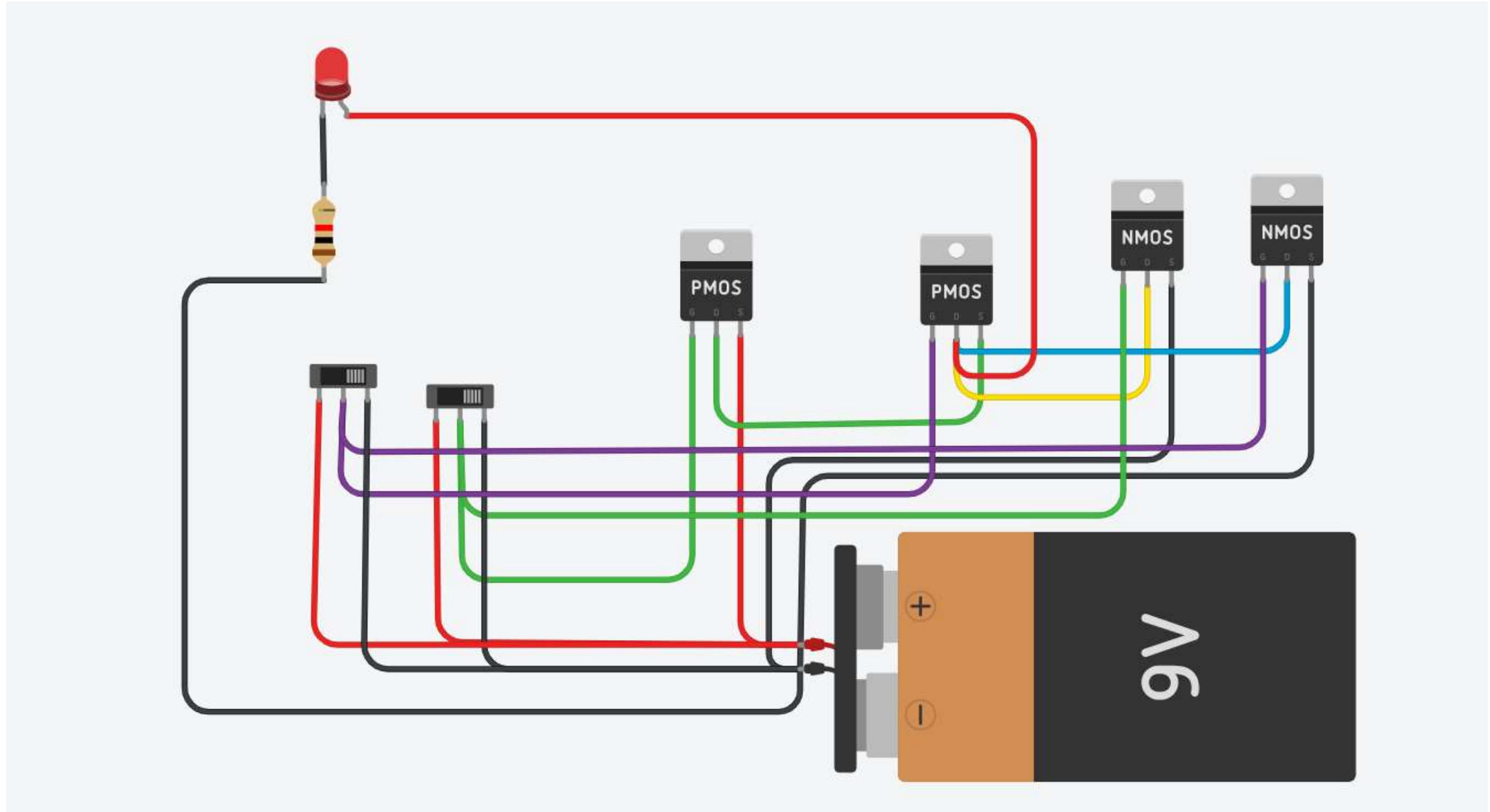
Name	Quantity	Component
T1 T2	2	pMOS Transistor (MOSFET)
T3 T4	2	nMOS Transistor (MOSFET)
BAT1	1	9V Battery
D1	1	Red LED
R1	1	1 kΩ Resistor
S2 S1	2	Slideswitch

CIRCUIT DIAGRAM

THE CIRCUIT IS A STATIC CMOS NOR GATE. THE TWO PMOS TRANSISTORS (T₁, T₂) ARE CONNECTED IN SERIES BETWEEN THE 9 V SUPPLY AND THE OUTPUT NODE, AND THE TWO NMOS TRANSISTORS (T₃, T₄) ARE CONNECTED IN PARALLEL BETWEEN THE OUTPUT NODE AND GROUND. EACH INPUT DRIVES THE GATE OF ONE PMOS AND ONE NMOS TRANSISTOR. THE OUTPUT NODE FEEDS THE LED THROUGH THE 1 K Ω CURRENT-LIMITING RESISTOR TO GROUND.

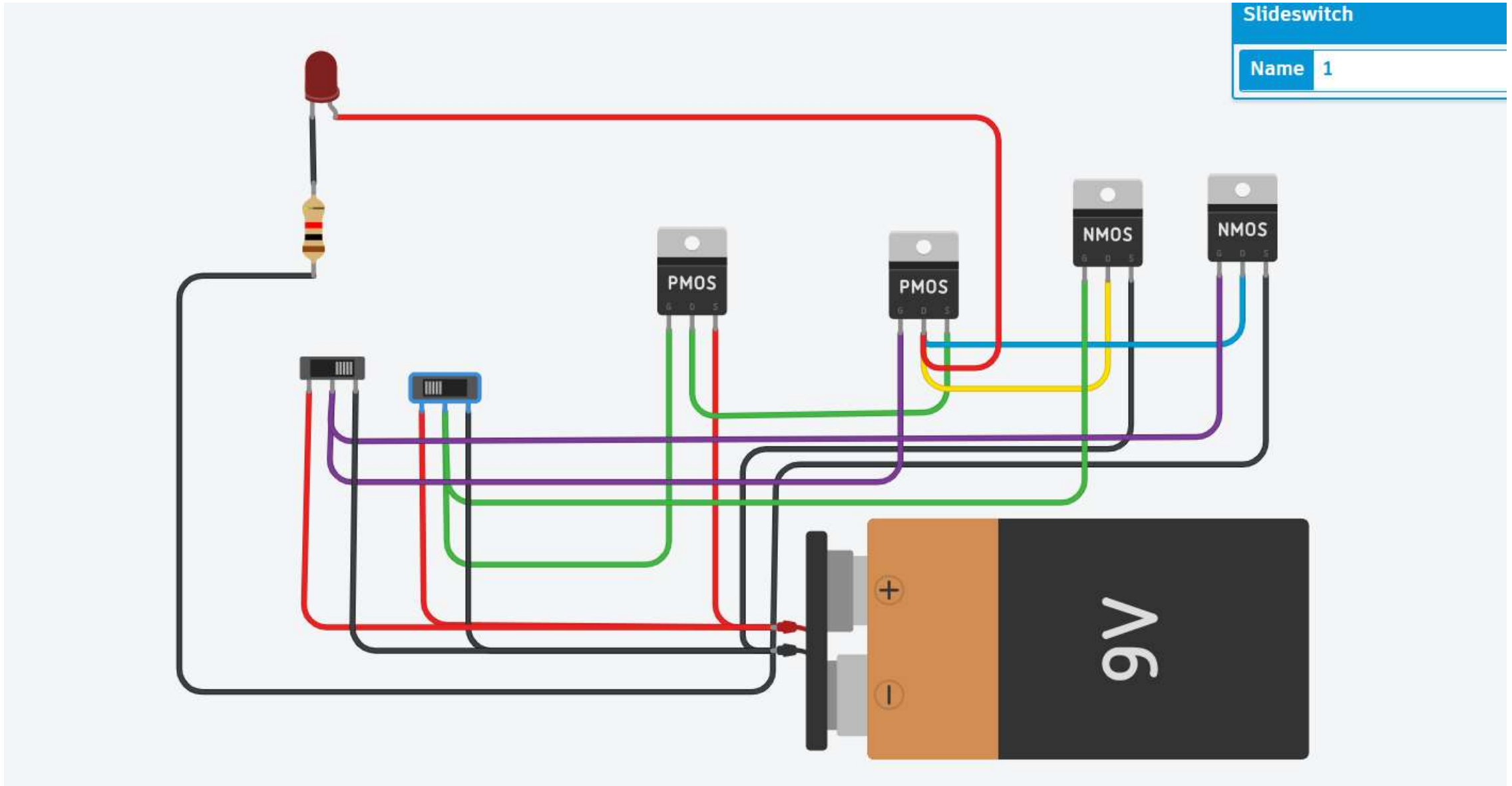


WHEN SWITCH 1-0 AND SWITCH 2-0 THE OUTPUT IS **1**



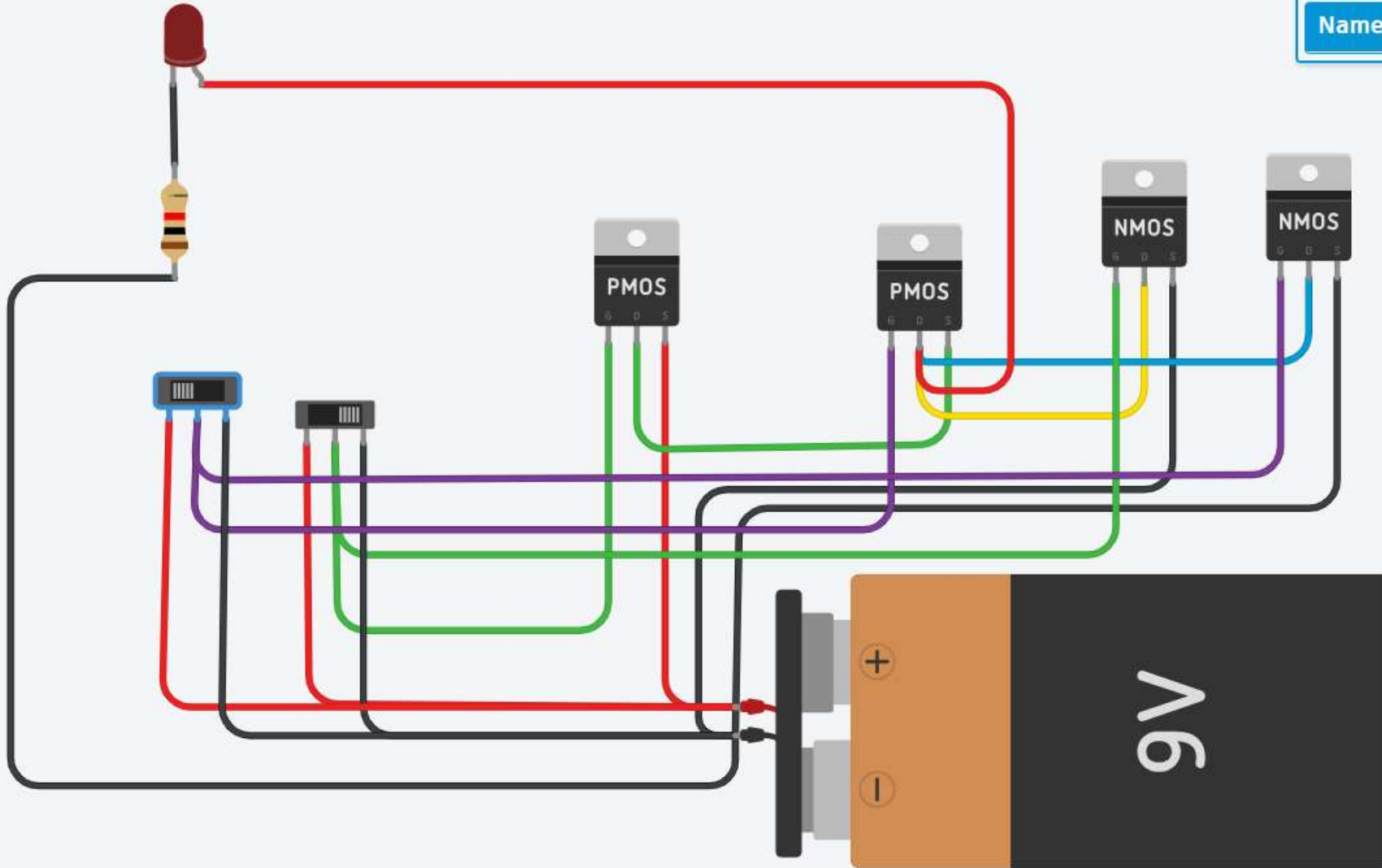
WE CAN SEE IN THE GIVEN FIGURE THAT THE LED IS **GLOW**

WHEN SWITCH 1-1 AND SWITCH 2-0 THE OUTPUT IS 0



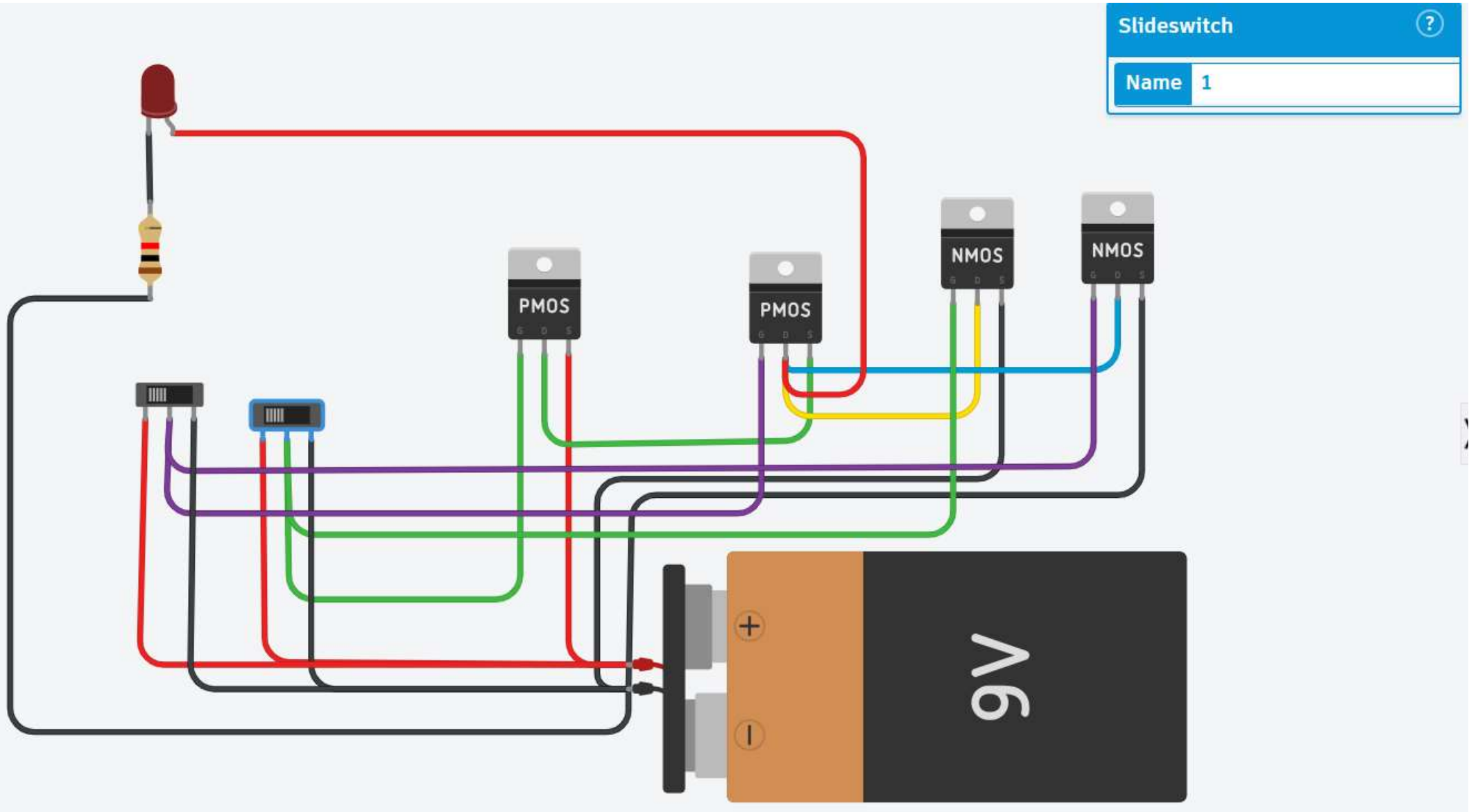
WE CAN SEE IN THE GIVEN FIGURE THAT THE LED IS NOT GLOW

Name 2



Name 2

WHEN SWITCH 1-1 AND SWITCH 2-1 THE OUTPUT IS 0



WE CAN SEE IN THE GIVEN FIGURE THAT THE LED IS NOT GLOW

Circuit working

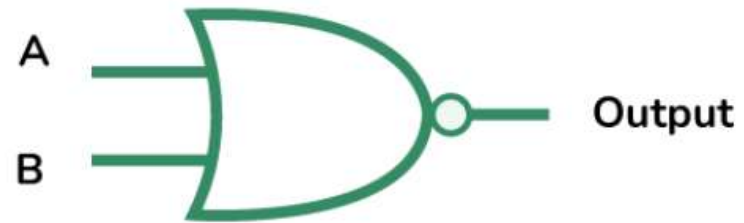
Each slide switch connects its gate line either to +9 V or to GND . A PMOS transistor conducts when its gate is LOW and an NMOS transistor conducts when its gate is HIGH, so the two networks are complementary: exactly one of them connects the output to a supply at any time.

- Pull-up network (series PMOS): Y is connected to +9 V only if both TRANSISTOR 1 and TRANSISTOR 2 conduct, i.e. $A = 0$ AND $B = 0$.

- Pull-down network (parallel NMOS): Y is connected to GND if TRANSISTOR 3 OR TRANSISTOR 4 conducts, i.e. $A = 1$ OR $B = 1$. ' ' Hence $Y = 1$ only when both inputs are 0, and $Y = 0$ otherwise, which is the NOR function: $Y = (A + B) ' = A \cdot B$.

TRUTH TABLE OF NOR GATE

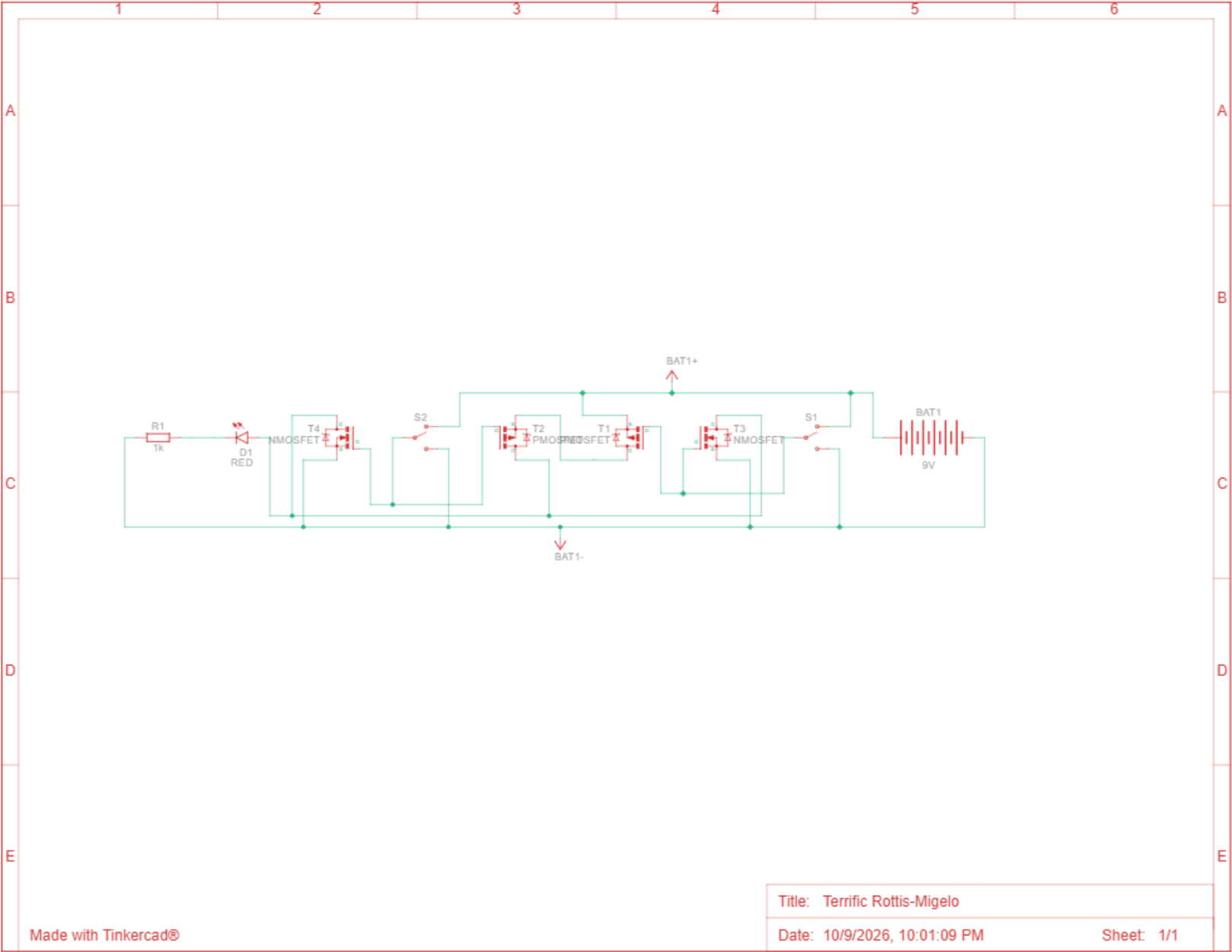
2- Input NOR Gate



Truth Table

Input A	Input B	$0 = (A + B)'$
0	0	1
0	1	0
1	0	0
1	1	0

SCHEMATIC DIAGRAM



CONCLUSION-

A two-input NOR gate was successfully constructed using two series PMOS and two parallel NMOS transistors and simulated in Tinkercad. The LED output confirmed the NOR truth table: the output is HIGH only when both inputs are LOW.

